

Cloud Recon Report

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Overview

Map UCI's cloud-hosted application assets reachable from the public Internet, starting from `uci.edu` and its subdomains, IPv4 only. Scope: discover → resolve → attribute to cloud provider → probe HTTP → flag risky exposures.

All numbers below come from running the four scripts in `scripts/`. Raw artifacts are in `data/`.

Methodology

Five iterative steps, each producing input for the next:

1. **Passive subdomain enumeration** — `subfinder -d uci.edu -all` (aggregates ~30 passive sources) plus a direct `crt.sh` Certificate Transparency query for `%.uci.edu`. Merge + dedupe.
2. **Internal-pattern filter** — drop obvious internal hostnames (cameras, VLAN gateways, workstation DHCP names, MAC-address-shaped hosts) by regex before resolving. Cuts query volume by ~46% without losing public-facing services.
3. **DNS resolution** — `dig @1.1.1.1 +short A` per subdomain, parallelized via `xargs -P 200`. IPv4 only.
4. **Cloud attribution** — match each unique IP against published CIDR ranges from AWS / GCP / Oracle / Cloudflare, with UCI's own ARIN-registered prefixes (`128.195/16` , `128.200/16` , `169.234/16` , `160.87/16`) hard-coded as fast paths. IPs not matched fall back to bulk `whois -h whois.cymru.com` ASN lookup; the holder string is keyword-classified (Microsoft→Azure, Salesforce, Akamai, ...). RFC1918 IPs get their own bucket.
5. **HTTP probing** — `curl -sk` per cloud-hosted subdomain, https-first then http fallback, capturing status code, `Server:` header, and HTML `<title>`. Run only on cloud-attributed hosts (≈ 2.9K) — on-prem UCI hosts are out of the assignment scope.

Iteration example in this run. Resolving `ad-rdpwgw-test.ad.uci.edu` from step 3 returns a CNAME chain, not a flat A record:

```
$ dig +short ad-rdpgw-test.ad.uci.edu
rd-gateway-dev.ad.uci.edu.
rd-gw-nlb-test12-ad477743bde42145.elb.us-west-2.amazonaws.com.
54.191.253.111
34.214.82.255
```

The intermediate ELB hostname `rd-gw-nlb-test12-...elb.us-west-2.amazonaws.com` is **not** a `*.uci.edu` cert SAN and neither `subfinder` nor `crt.sh` could surface it on their own — only a DNS follow-up can. From that single chain an attacker learns three things for free: (a) the deployment uses a Network Load Balancer (NLB) rather than ALB, (b) it lives in `us-west-2`, and (c) the artifact suffix `test12` suggests this is the 12th iteration of an internal RDP-gateway test environment. We also tried feeding back TLS-cert SANs (`openssl s_client | x509 -text | grep DNS`): the shared cert on `app.commencement.uci.edu` covers 151 hostnames in one shot, but in this run all 151 were already in `subdomains.txt` because `crt.sh` had indexed the same CT entry — a useful **negative** finding that validates `crt.sh` coverage rather than uncovering new names. The two iteration channels (CNAME chase vs cert-SAN feedback) thus produced different value: one revealed AWS-side topology, the other confirmed enumeration completeness.

Tools used

Tool	Purpose	Why chosen	Drawback
subfinder	Aggregate passive subdomain sources	Single CLI fronts ~30 APIs, free + fast	Pulls a lot of internal/DHCP noise — needs filtering
crt.sh JSON	Certificate Transparency hostnames	Authoritative ("a cert was issued")	Misses internal-only hosts; rate-limits
dig +short	DNS resolution	Built-in, scriptable	Slow at scale without parallelism
xargs -P	Parallelism	No new dependencies	Coarse — no per-host backoff
AWS/GCP/Oracle/Cloudflare CIDR JSONs	Cloud attribution	Vendor-published, current	Azure has no stable public JSON —

Tool	Purpose	Why chosen	Drawback
			covered via ASN fallback
Team Cymru whois.cymru.com	Bulk ASN lookup	One TCP session resolves all leftover IPs in seconds	Holder strings are free-form — needs keyword classifier
curl + <title> grep	HTTP fingerprint	Trivial, no httpx / nuclei dependency	No TLS cert inspection, no tech-stack detection beyond Server:

Considered but skipped: amass (overkill for assignment scope), httpx / nuclei (active vuln scanning is out of scope), DNS bruteforce wordlists (passive sources were sufficient).

Alternatives considered (pros/cons summary): amass vs subfinder — amass adds active brute-force on top, 10x slower and noisier, marginal coverage gain for a public university domain that's already well-indexed; subfinder passive-only was sufficient. Censys / SecurityTrails APIs vs crt.sh — paid tiers cover the same CT data plus historical DNS, but crt.sh is free, JSON-queryable, and authoritative for "a cert was issued". dnsx / massdns vs dig + xargs — order-of-magnitude faster on large input, but external dependency; dig finished 89K in 3 min, fast enough. httpx / nuclei vs curl — better fingerprinting + vuln signatures, but nuclei actively probes for vulns, which is outside the read-only scope of this assignment.

Findings

Enumeration funnel

Stage	Count
Raw subdomains from subfinder -all	164,237
From crt.sh (CT logs)	2,061
Merged + deduped	164,619
After internal-pattern filter	89,375

Stage	Count
Resolved to ≥ 1 IPv4 A record (unique subs)	59,622
Unique IPv4 addresses	56,264

Cloud provider distribution

By **(subdomain, IP) pairs**, totaling 62,655:

Provider	Pairs	Notes
UCI on-prem (UCnet / UCI Health)	56,620	128.195/16, 128.200/16, 169.234/16, 160.87/16 + ASN-matched
AWS	4,785	mostly us-west-2 (Oregon) + us-east-2 (Ohio); a few in us-gov-west-1
RFC1918 leaked into public DNS	282	180 unique internal IPs (all 10.65.x.x — UCI AD subnet)
DigitalOcean	274	*.sandbox.lib.uci.edu (library sandboxes)
Azure	239	matched via ASN fallback (no published JSON)
Other (SaaS / 3rd-party)	169	Automattic (WordPress.com), Squarespace, Weebly, SDSC, Cox, Proofpoint, Sucuri, Pantheon, Rackspace
Fastly	126	CDN edge
Cloudflare	93	CDN edge
GCP	46	small footprint
Salesforce	8	community / CRM
Oracle	7	small footprint
Akamai	6	CDN

Unique **cloud-hosted subdomains** (everything except UCI on-prem / RFC1918): **2,909**.

AWS region breakdown (parsed from the AWS IP-ranges JSON, by (subdomain, IP) pairs):

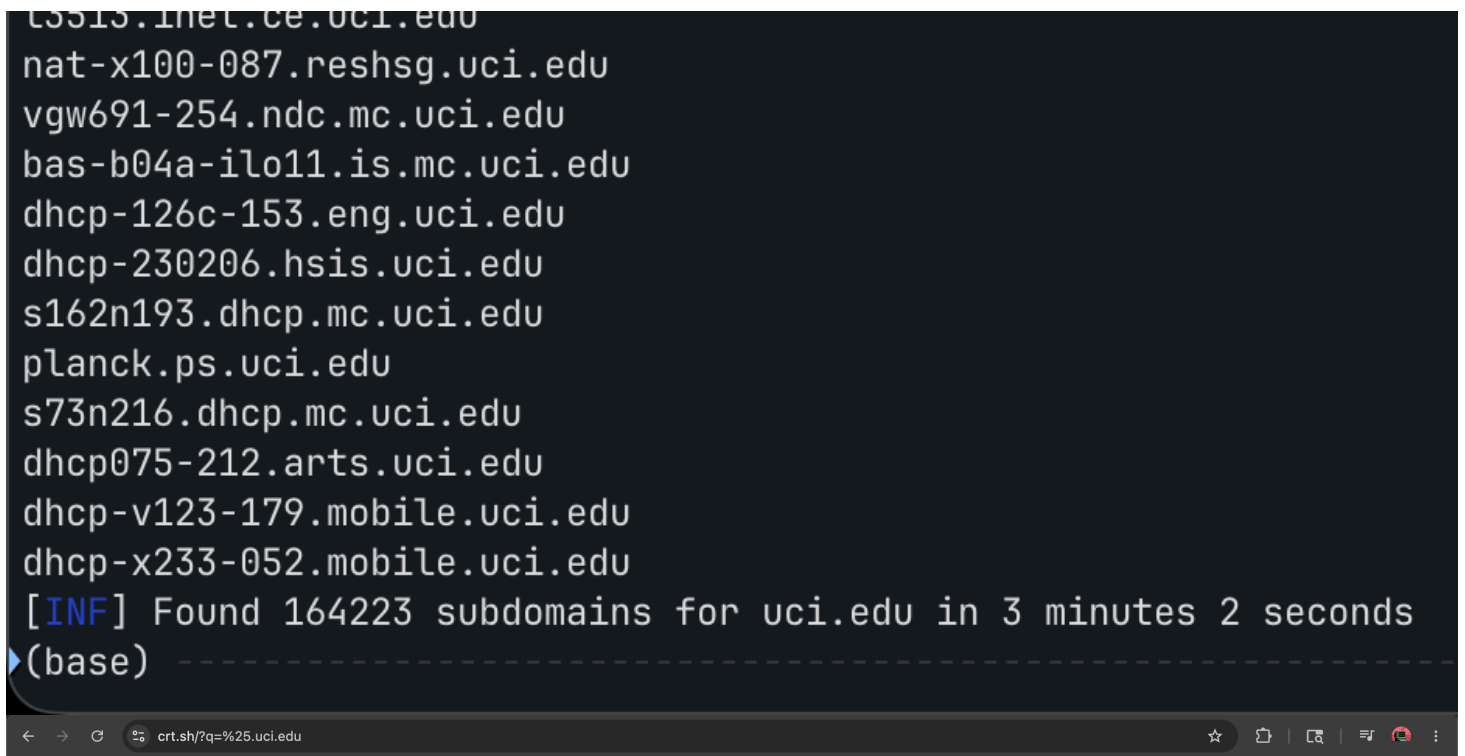
Region	Pairs
us-west-2 (Oregon — primary)	2,682
us-east-2 (Ohio)	1,750
GLOBAL (S3 / CloudFront / IAM endpoints)	240
us-east-1 (N. Virginia)	64
us-west-1 (N. California)	40
us-gov-west-1 (GovCloud)	3
(uncategorized, ASN-bucket fallback)	6

The 6 uncategorized rows hit AWS via the Team Cymru ASN lookup but did not match any region prefix in the AWS IP-ranges JSON snapshot — typically newly-assigned blocks not yet in the published file, or rows where the JSON's `region` field was missing. UCI is heavily centered on `us-west-2` with `us-east-2` as a secondary, suggesting an active-passive or geo-distributed setup. The `us-gov-west-1` slice is small but covered separately in the risk section.

Attack surface summary

UCI's public-Internet **cloud attack surface** breaks down as $\approx 2,909$ unique cloud-hosted subdomains across at least 11 distinct providers, dominated by AWS (~75% of cloud subdomains) with a meaningful long tail (Azure, GCP, Cloudflare, Fastly, DigitalOcean, Oracle, Akamai, Salesforce, plus ~10 SaaS tenancies). On top of that, ~180 RFC1918 records leak internal infrastructure names into public DNS, and the AWS footprint spans multiple regions including `us-gov-west-1` (GovCloud). The detailed risky exposures are enumerated in the next section.

dhcp-x212-104.hs.uci.edu
kenya.ics.uci.edu
s113n22.surg.mc.uci.edu
ieee-club-x132066.eecs.uci.edu
oaacs134-135.oaa.uci.edu
s127n105.dhcp.mc.uci.edu
tibco-db-d-1.ad.uci.edu
b88192.ad.uci.edu
dhcp-148a-51.bio.uci.edu
ldap-prod6.oit.uci.edu
cts.testing12.oit.uci.edu
dhcp-203250.bio.uci.edu
dhcp-v030-112.mobile.uci.edu
dhcp-v094-056.mobile.uci.edu
s139n42.dhcp.mc.uci.edu
blog.odit.uci.edu
dhcp-v122-205.mobile.uci.edu
dhcp-v168-184.me.reshsg.uci.edu
dhcp-v195-039.cv.reshsg.uci.edu
s94n62.dhcp.mc.uci.edu
www.angagepuci.bio.uci.edu
dhcp-v033-013.mobile.uci.edu
dhcp-v200-016.cv.reshsg.uci.edu
s150n221.dhcp.mc.uci.edu
s227n88.dhcp.mc.uci.edu
vcv077151.vpn.uci.edu
dhcp-126c-156.eng.uci.edu
s225n178.dhcp.mc.uci.edu
webdev.nacs.uci.edu
www.training.uci.edu
dhcp-v231-206.pv.reshsg.uci.edu
mst40.ps.uci.edu
za02.is.mc.uci.edu
dhcp-v057-014.mobile.uci.edu
simulate.ps.uci.edu
726-vl101.ucinet.uci.edu
dhcp-128-195-254-220.studentcenter.uci.edu
dhcp-v024-175.mobile.uci.edu
dhcp-v081-249.mobile.uci.edu
17517.inet.cs.uci.edu



crt.sh Identity Search



Group by Issuer

Criteria Type: Identity Match: ILIKE Search: 'uci.edu'

Sorry, your search results have been truncated.
It is not currently possible to sort and paginate large result sets efficiently, so only a random subset is shown below.
Please retry your search with [expired certificates excluded](#).

Certificates	crt.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities	Issuer Name
	3488050431	2020-10-10	2016-09-22	2019-09-22	ad-cp-live.ad.uci.edu	ad-cp-live.ad.uci.edu	C=US, ST=MI, L=Ann Arbor, O=Internet2, OU=InCommon, CN=InCommon RSA Server CA
	2387344252	2020-01-29	2005-09-24	2007-10-26	www.humanities.uci.edu	www.humanities.uci.edu	C=ZA, ST=Western Cape, L=Cape Town, O=Thawte Consulting cc, OU=Certification Services Division, CN=Thawte Premium Server CA, emailAddress=premium-server@thawte.com
	2387300885	2020-01-29	2008-12-12	2010-01-12	sdcs.unx.uci.edu	sdcs.unx.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2387291304	2020-01-29	2006-09-07	2011-10-07	scandocs.ddm.uci.edu	scandocs.ddm.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2387243600	2020-01-29	2007-07-17	2009-09-15	www.fs.uci.edu	www.fs.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2387203592	2020-01-29	2007-10-02	2009-10-02	catalyst.merage.uci.edu	catalyst.merage.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2386824540	2020-01-29	2008-02-25	2010-03-27	www.merage.uci.edu	www.merage.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2386677307	2020-01-29	2007-02-21	2008-03-23	catalyst.gsm.uci.edu	catalyst.gsm.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2386661244	2020-01-29	2006-10-11	2008-12-10	apps.adcom.uci.edu	apps.adcom.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2386547927	2020-01-28	2007-02-13	2008-02-14	www.merage.uci.edu	www.merage.uci.edu	C=US, O=Equifax Secure Inc., CN=Equifax Secure Global eBusiness CA-1
	2380528413	2020-01-26	2010-11-08	2011-11-03	hr.healthcare.uci.edu	hr.healthcare.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380532024	2020-01-26	2010-11-08	2011-11-03	hrconnect.healthcare.uci.edu	hrconnect.healthcare.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380528418	2020-01-26	2011-01-06	2013-02-03	fdx.cap.uci.edu	fdx.cap.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380525894	2020-01-26	2010-07-29	2011-10-25	www.humanities.uci.edu	www.humanities.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380511110	2020-01-26	2010-11-08	2011-11-03	careers.healthcare.uci.edu	careers.healthcare.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380502834	2020-01-26	2010-08-31	2011-08-16	info-globaldesk.hs.uci.edu	info-globaldesk.hs.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380492488	2020-01-26	2010-12-28	2013-01-18	VCDDEAN.HS.uci.edu	VCDDEAN.HS.uci.edu	C=US, O=Thawte, Inc., CN=Thawte SSL CA
	2380436977	2020-01-26	2011-03-02	2011-11-02	catalyst.merage.uci.edu	catalyst.merage.uci.edu	C=US, O=GeoTrust Inc., OU=Domain Validated SSL, CN=GeoTrust DV SSL CA
	2380347683	2020-01-26	2010-11-05	2011-11-05	unex.uci.edu	unex.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)10, CN=VeriSign Class 3 Secure Server CA - G3
	2380299675	2020-01-26	2010-01-05	2011-01-05	exchange.uci.edu	ad.uci.edu autodiscover.ad.uci.edu autodiscover.exchange.uci.edu autodiscover.law.uci.edu cas.ad.uci.edu exchange.uci.edu law.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)10, CN=VeriSign Class 3 Secure Server CA - G3
	2380156818	2020-01-26	2010-07-28	2011-07-28	nw.reg.uci.edu	nw.reg.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)09, CN=VeriSign Class 3 Secure Server CA - G2
	2380038670	2020-01-26	2010-01-28	2011-01-28	summer.uci.edu	summer.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)09, CN=VeriSign Class 3 Secure Server CA - G2
	2380014343	2020-01-26	2009-10-07	2010-11-05	unex.uci.edu	unex.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)09, CN=VeriSign Class 3 Secure Server CA - G2
	2379982911	2020-01-26	2009-05-26	2010-06-16	ring.reg.uci.edu	ring.reg.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)09, CN=VeriSign Class 3 Secure Server CA - G2
	2379977720	2020-01-26	2009-07-14	2011-09-10	www.reg.uci.edu	www.reg.uci.edu	C=US, O=VeriSign, Inc., OU=VeriSign Trust Network, OU=Terms of use at https://www.verisign.com/rpa (c)09, CN=VeriSign Class 3 Secure Server CA - G2

HTTP probe summary

Of the 2,909 unique cloud-hosted subdomains, **2,354** responded to an HTTP or HTTPS request (others timed out, refused, or had no listening service):

Status class	Count	% of responders
2xx OK	1,169	49.7 %
3xx redirect (mostly to SSO / login)	798	33.9 %
4xx / 5xx	387	16.4 %

Top Server: headers seen:

Server	Count
openresty	733
Apache (generic)	627
Apache/2.4.62 (Rocky Linux) OpenSSL/3.5.1	166
nginx	144
Microsoft-IIS/10.0	111
awselb/2.0	77
cloudflare	75
Apache/2.4.58 (IUS) OpenSSL/1.0.2k-fips PHP/7.4.33	42
AmazonS3	42
Microsoft-HTTPAPI/2.0	38
cPanel	19

Worth noting: an Apache build advertising `PHP/7.4.33` shows up 42 times — PHP 7.4 has been end-of-life since November 2022, so any application still running on it should have a migration plan.

Detected applications. Server-header counts alone overcount stacks like [ASP.NET](#) (an IIS server can host static files or any backend), so for the IIS subset I ran a follow-up HEAD probe (`curl -skI`) capturing `X-Powered-By` and `X-AspNet-Version` to confirm the application layer:

Application	Count	Detection signal
ASP.NET (IIS-backed)	92	<code>X-Powered-By: ASP.NET</code> confirmed via follow-up HEAD probe; 34 are <code>autodiscover.* Exchange→O365</code> redirectors, the remaining 58 are first-party UCI apps

Application	Count	Detection signal
WordPress	38	<title> patterns + wp- paths in redirects
cPanel (Site Publisher)	38	Server: cPanel or "Powered by cPanel" in body
Tomcat (behind Apache + mod_jk)	22	Server: Apache/... mod_jk/...

Surprising / risky exposures

Each finding is tagged with a severity reflecting blast radius × exploitability for an external attacker:

High = direct path to UCI data or services; **Medium** = enables follow-on attacks or reflects systemic hygiene gaps; **Low** = informational / context-setting.

1. RFC1918 internal IPs leaked into public DNS [Medium]

180 unique 10.65.x.x addresses are returned by public DNS for names like ad-rds-lic3.ad.uci.edu , ad-sn-mid04.ad.uci.edu , ad-oit-rd-jump1.ad.uci.edu , adm-d-web01.oars.uci.edu . The IPs themselves aren't routable from the Internet, but the records disclose internal Active Directory infrastructure (license servers, RDP gateways, ServiceNow MID servers, dev databases) — useful intel for any attacker who has already gotten a foothold on campus and wants to pivot. Classic split-horizon DNS that isn't actually split.

```
$ awk -F, '$3=="RFC1918 (leaked)'" data/cloud_hosts.csv | head -3
ad-adrap.ad.uci.edu,10.65.88.5,RFC1918 (leaked),...
ad-rds-lic3.ad.uci.edu,10.65.77.7,RFC1918 (leaked),...
ad-sn-mid04.ad.uci.edu,10.65.92.36,RFC1918 (leaked),...
```

```
(base)
~/Courses/SWE267P/assignment4 (main*) » dig +short ad-rds-lic3.ad.uci.edu
10.65.77.7
(base)
~/Courses/SWE267P/assignment4 (main*) » dig +short ad-sn-mid04.ad.uci.edu
10.65.92.36
(base)
~/Courses/SWE267P/assignment4 (main*) » dig +short ad-oit-rd-jump1.ad.uci.edu
10.65.70.10
(base)
```

2. UCI hosts Time Reporting System (TRS) for UCB, UC Merced, UC Law SF on its AWS account [High]

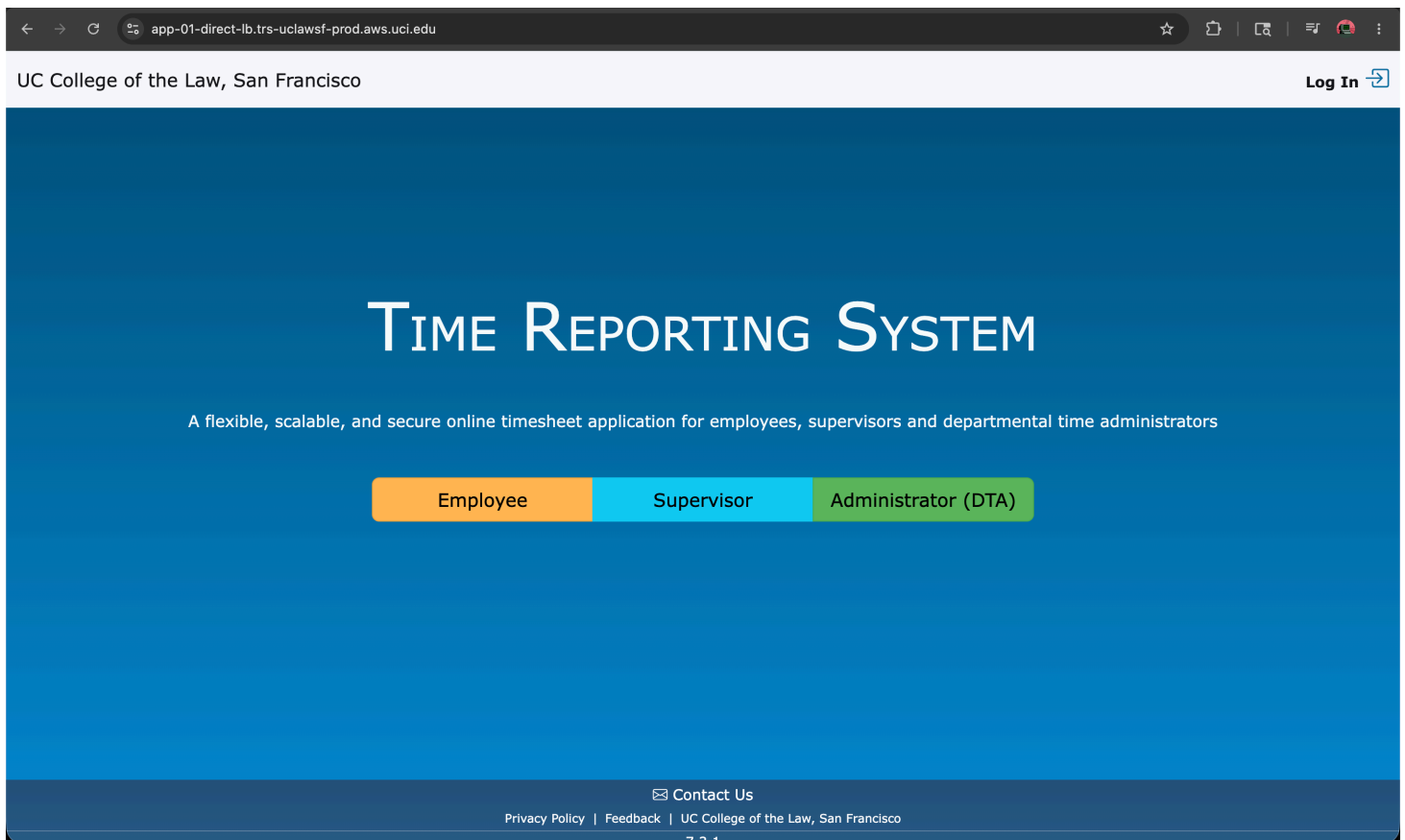
*.aws.uci.edu reveals multi-environment, multi-campus deployments:

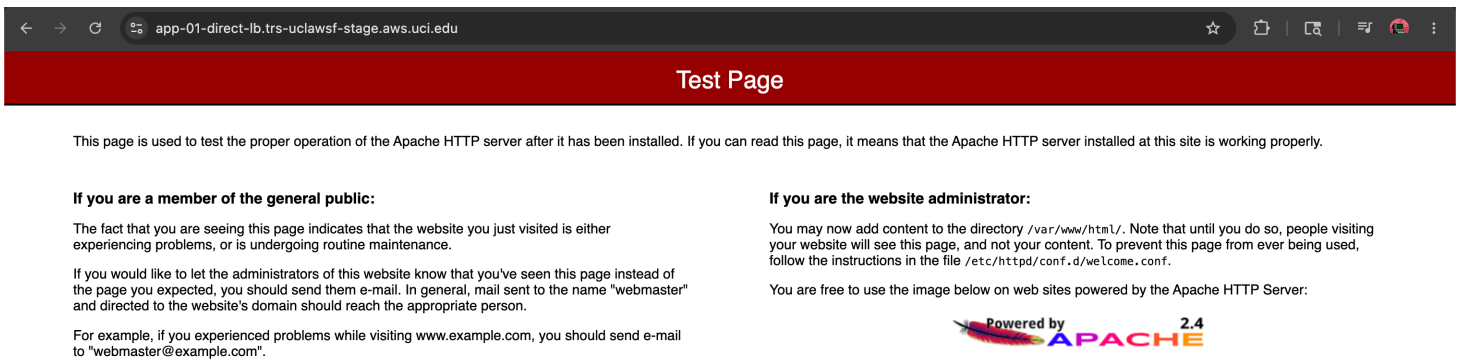
trs-ucb-{dev,stage,prod}.aws.uci.edu	– UC Berkeley TRS
trs-ucm-{stage,prod}.aws.uci.edu	– UC Merced TRS
trs-uclawsf-{stage,prod}.aws.uci.edu	– UC Law San Francisco TRS

The *-prod hosts respond 200 OK with title

Index – UC <Campus> Time Reporting System – TRS . The matching *-stage and *-dev load balancers respond 403 with the default Test Page for the Apache HTTP Server or

It works! Apache httpd — i.e., the LB is reachable but no app is bound, which both leaks the existence of the environment and suggests the deployment is incomplete. Notable because (a) the naming standard publishes the entire dev/stage/prod topology and (b) UCI is a single tenant for at least three other UC campuses' payroll-adjacent workflows, which is a cross-campus blast radius worth knowing about.





3. Default web-server pages on real *.uci.edu hosts [Medium]

Beyond TRS, ~30+ hosts return Apache or nginx default pages, e.g.:

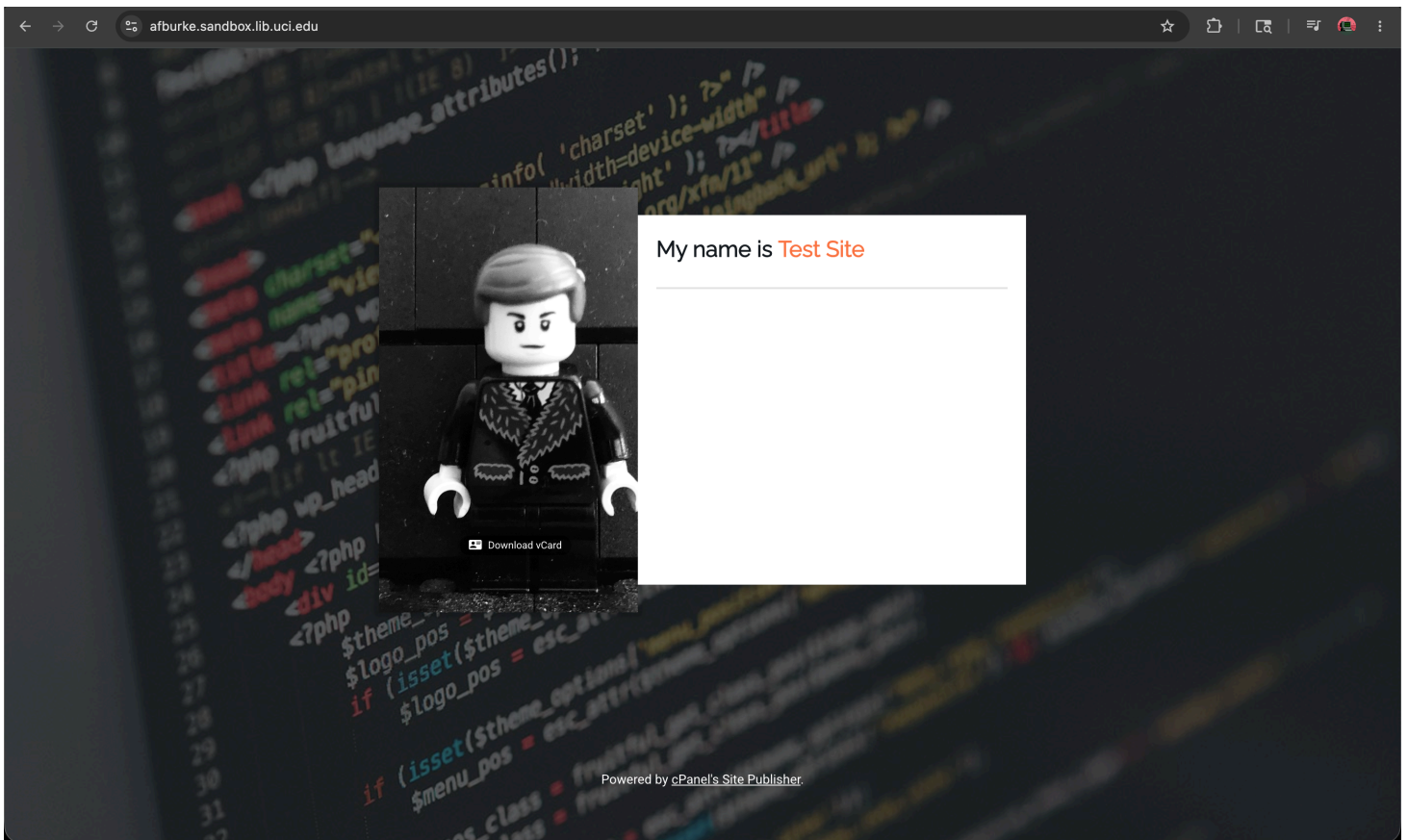
- apps.athletics.uci.edu (prod) — Apache test page
- prod.apps.athletics.uci.edu , prod.students.athletics.uci.edu , prod.data.athletics.uci.edu — Apache test page
- kualidocs.oit.uci.edu — Apache test page
- app.commencement.uci.edu — Rocky Linux test page
- bw.hs.uci.edu — nginx "This is the default server vhost"

These hosts are bound in DNS, certificate-issued, and live, but the back-end is either offline or misconfigured. From an attacker's view this is interesting because the host is "warm" and could host shadow content; from an operator's view it suggests stale deployments and DNS records that outlived the application.



4. *.sandbox.lib.uci.edu — personal sandboxes on DigitalOcean [Medium]

274 *.sandbox.lib.uci.edu records point at DigitalOcean droplets, named after individuals (afburke, alice, dailycey, fenai, kehan, ...). Three patterns are visible in the probe data: live "UCI Libraries Digital Sandbox Service" Apache, plain Test Site Apache (default install never configured), and autodiscover.<user>.sandbox.lib.uci.edu MX-style records that return Microsoft Outlook autodiscover 302s. Patching, MFA, and lifecycle of these droplets are almost certainly inconsistent across users, and DNS records persist after droplet teardown unless explicitly cleaned up.



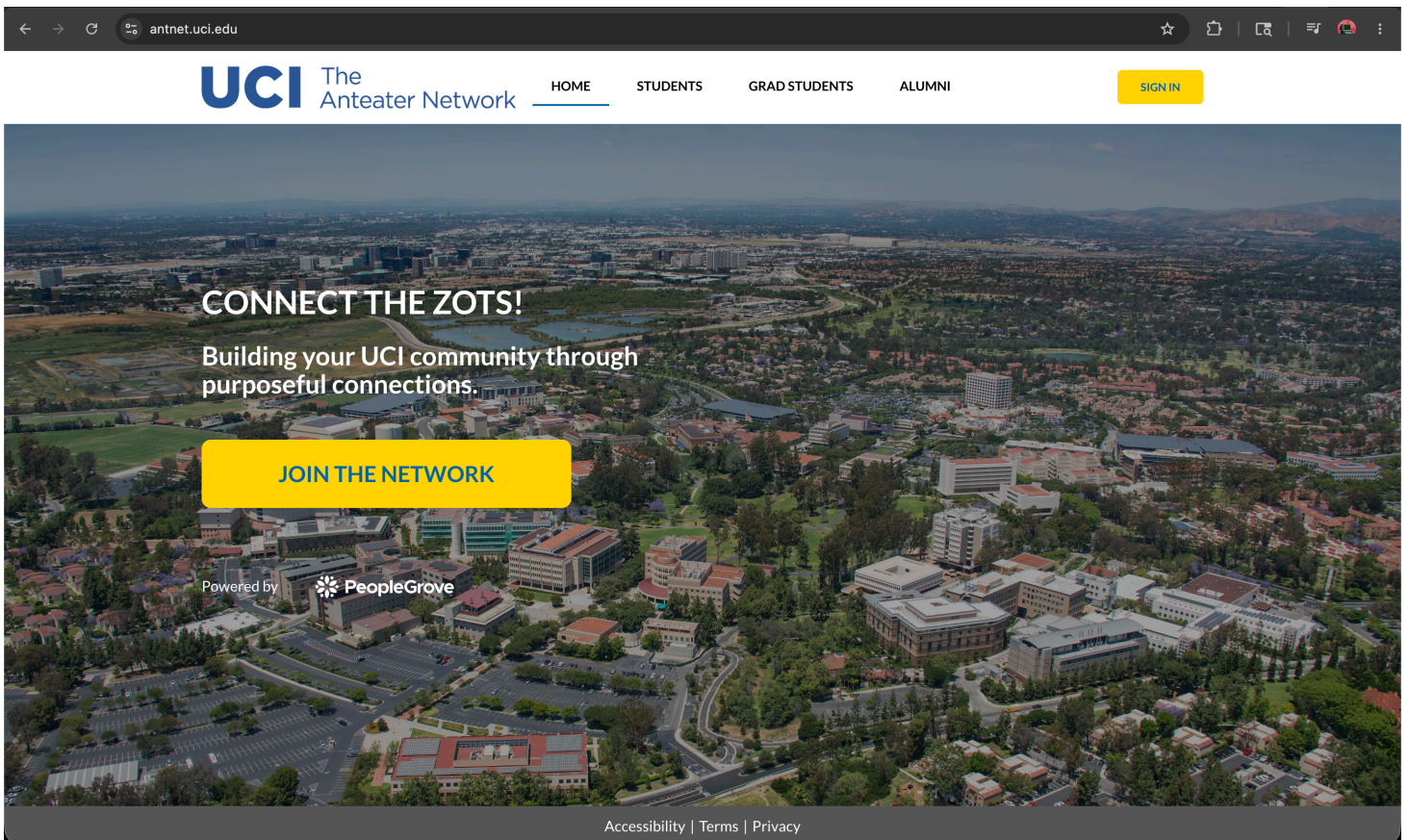
5. GovCloud presence (us-gov-west-1) [Low]

engage.police.uci.edu resolves to three AWS GovCloud IPs (3.30.76.196 , 3.31.3.176 , 52.222.14.14). GovCloud is the AWS region tier reserved for CJIS / FedRAMP-class data — consistent with this being the UC Irvine Police Department's external engagement portal, which would handle reporting / records workflows subject to law-enforcement data rules. Worth flagging not because it's "exposed" but because it implies a high-sensitivity workload whose audit scope (CJIS, possibly FedRAMP-Moderate) should be checked explicitly; any future expansion into GovCloud should pass through the same governance.

6. Long-tail SaaS surface [Medium]

*.uci.edu DNS resolves into Salesforce, Squarespace, Weebly, Automattic (wordpress.com), Pantheon, Rackspace, Sucuri, and even an SDSC ASN. Each is a separate tenancy that someone in UCI provisioned and presumably owns; the central security team likely has no inventory of all of them.

antnet.uci.edu	→ PeopleGrove (career-services SaaS)
*.healthcare.uci.edu (some)	→ cloudflare
research.test-pantheon.bio.uci.edu	→ Pantheon (staging — name says so)



Recommendations

Tagged by priority: **[P1]** **[P2]** **[P3]**

1. **[P1] Decouple internal DNS from public DNS.** Stop publishing `*.ad.uci.edu` -style records that resolve to RFC1918 addresses. At minimum move these into a split-horizon view that doesn't answer to public resolvers.
2. **[P1] Sweep for default web-server pages.** A trivial weekly probe that flags titles like "Test Page for the Apache HTTP Server", "It works!", or "Welcome to nginx" on any `*.uci.edu` host would surface stale / broken deployments before someone else does.
3. **[P2] Adopt a naming standard that does not disclose environment.** Avoid `*-dev`, `*-stage`, `*-prod` in public DNS. If dev/stage must be reachable, gate them behind SSO + IP allowlists rather than just relying on an attacker not knowing the hostname.
4. **[P2] Catalogue and govern SaaS tenancies.** Require central registration for any `*.uci.edu` CNAME pointing at a third-party SaaS. Pair every external tenancy with a UCI owner of record and a renewal review.
5. **[P2] Lifecycle library/researcher sandboxes.** For `*.sandbox.lib.uci.edu` and similar personal-droplet patterns, define a 90/180-day expiry, require MFA + a patching baseline, and auto-clean DNS records on droplet teardown.

6. **[P3] Build a continuous external asset inventory.** Re-run a pipeline like the one in `scripts/` weekly. Diff the cloud-attribution result to flag new third-party SaaS tenancies and new cloud accounts.
7. **[P3] Subscribe to Certificate Transparency for `uci.edu`.** Newly-issued certs (= newly-deployed services) should trigger an alert to the central security team. crt.sh and `cert-monitor` -style tools make this cheap.

Challenges & lessons

The biggest practical issue was scale: `subfinder` returned ~165K hostnames, the majority of which are individual workstations, cameras, and VLAN gateways scraped from PTR-record sources. Without the pattern filter and a high-parallelism resolver the resolution step would have taken close to an hour. The simplest mitigation was a small set of regex exclusions plus `xargs -P 200` against a public resolver (`1.1.1.1`); end-to-end resolution then took under three minutes.

Cloud attribution was a different shape of problem: AWS, GCP, Oracle, and Cloudflare publish stable IP-range JSONs, but Microsoft Azure does not (the link is behind a JS-gated click-through). The cheap and reliable workaround was to fall back to Team Cymru's bulk WHOIS server for any IP not matched by a published range, then keyword-classify the holder string. That covered Azure, on-campus prefixes the published ranges miss, and every SaaS provider as a bonus.

The other notable surprise was how much can be inferred from naming alone — several of the findings above (RFC1918 leaks, dev/stage topology, multi-campus TRS) require no probing at all, only DNS reads. That argues for treating DNS hygiene as a security control, not just a networking concern.

Appendix — reproducing this report

```
cd assignment4
bash scripts/01_enum.sh uci.edu # → data/subdomains*.txt
bash scripts/02_resolve.sh data/subdomains_filtered.txt # → data/resolved.csv
python3 scripts/03_cloud_attr.py data/resolved.csv # → data/cloud_hosts.csv
# Filter to cloud-attributed only:
awk -F, 'NR==1 || ($3!="UCI on-prem" && $3!="UCI on-prem / UCnet" && $3!="RFC1918 (leak
data/cloud_hosts.csv > data/cloud_hosts_external.csv
bash scripts/04_probe.sh data/cloud_hosts_external.csv # → data/probe.csv
```

Spot-checks:

```
# Confirm one AWS attribution
```

```
awk -F, '$3=="AWS"{print $2; exit}' data/cloud_hosts.csv | xargs whois | grep -i amazon
```

```
# Confirm one RFC1918 leak
```

```
dig +short ad-rds-lic3.ad.uci.edu
```

```
# Show all default Apache test pages found
```

```
awk -F, 'NR>1 && $4~/[Tt]est Page for the Apache|[Ii]t [Ww]orks|default server vhost/'
```

```
# Count TRS multi-campus hosts
```

```
awk -F, 'NR>1 && $1~/trs-(ucb|ucm|uclawsf)-/' data/probe.csv | wc -l
```